

AKB Additional Benchmark Baseline Validation

Phase 1 Task Specification

Goal 快速确定 SciFact / SciFact-Open 是否适合作为 AKB 的补充实验平台，并验证代表性 baseline 能否在现有公开实现上稳定跑通、输出官方指标。第一阶段只做“可运行性 + 指标可得性”验证，不做完整系统集成。

Execution summary

- 主数据集冻结为 SciFact 与 SciFact-Open；ResearchClawBench 延后到 agent/harness 补充实验。
- 必须完成：SciFact 的强 verifier baseline；SciFact-Open 的官方 evaluator sanity check + 一个可重跑的 open-domain retrieval baseline。
- 优先使用作者公开 checkpoint / prediction；禁止为了第一阶段重新训练大型模型。
- 单个 baseline 若在约半天内仍无法进入推理/评估阶段，记录阻塞原因并停止，不继续修旧依赖。
- 最终产物是代码、命令、环境记录和指标表，保证下一阶段可以把 AKB 模块替换进同一评测链路。

Recommended total effort: 8-12 focused hours

1. Project Context and Phase Boundary

当前主线系统以 ScholarQABench 为主要 benchmark 搭建 AKB/科学知识库雏形。本任务负责额外 benchmark 的快速验证：优先寻找具备真实科学语料、claim-evidence 关系、验证/冲突判别需求的数据集，并找到可以作为 AKB 对照的现成方法。

第一阶段的重点不是证明 AKB 优于 baseline，而是把“数据 -> baseline -> evaluator -> 指标”的链路跑通。只要链路稳定、指标可复现，即可进入下一阶段。

Scope lock P0 = SciFact + SciFact-Open。P1 = MetaSyn / SciQA-ORKG 仅在 P0 完成后做快速可用性扫描。ResearchClawBench、完整 agent harness、知识库增量更新实验均不进入本阶段。

1.1 Success criteria

- SciFact：至少 1 个强 verifier (MultiVerS) 在公开 dev split 上完成推理并得到官方 evaluator 输出。
- SciFact：若 VeriSci 官方 pipeline 可在时间预算内启动，再补 1 组 full-pipeline 指标；否则只记录阻塞，不影响阶段完成。
- SciFact-Open：官方提供的 predictions 能被 run_eval.py 正常评估，证明数据和 evaluator 完整。
- SciFact-Open：在完整 500K corpus 上建立一个简单 retrieval baseline，并输出正式 Recall@K / repository evaluator 支持的 retrieval 指标。
- 可选：将 retrieval top-K 转成 MultiVerS 所需 doc_ids，完成小规模或全量 end-to-end verifier smoke run。
- 所有成功/失败尝试均有命令、commit hash、环境、输出路径和简短结论。

1.2 Explicit non-goals

- 不重新训练 MultiVerS、+VeriRel 或其他大型 verifier/reranker。
- 不为了追论文 SOTA 数字做超参数搜索、多随机种子、交叉验证或大规模 error analysis。
- 不设计完整 AKB schema、confidence/wealth 机制或 provenance graph。
- 不做 LLM agent tool loop；如果某 baseline 必须依赖 agent/harness 才能运行，则移入后续阶段。
- 不把 corpus_candidates (~12K) 上的 retrieval 结果当作 SciFact-Open 正式结果。

2. Dataset Plan

Priority	Dataset	Role in Phase 1	Data / Scale	Required output
P0	SciFact	快速验证 scientific claim verification baseline；最容易得到标准指标。	Public train/dev/test claims; test labels hidden; corpus.jsonl.	MultiVerS dev metrics; optional VeriSci pipeline metrics.
P0	SciFact-Open	开放域科学语料检索 + claim verification；更接近后续知识库实验。	279 claims; full corpus = 500K S2ORC abstracts; candidate subset ~12K only for smoke/debug.	Official evaluator sanity result + full-corpus retrieval metrics.
P1	MetaSyn / SciQA-ORKG	仅作为后续候选，检查是否值得接入。	Do not download/prepare unless P0 finishes early.	1-page note: task, data access, evaluator, baseline availability.
Later	ResearchClawBench	Agent/harness 补充实验。	40 research tasks / multi-domain.	No action in Phase 1.

2.1 Important dataset constraints for the coding agent

- SciFact 的 test gold 不公开，因此本地可验证结果以 dev 为主；不要把 test prediction 当作可本地评分结果。
- SciFact-Open 的 data/corpus.jsonl 是正式 500K abstract corpus；data/corpus_candidates.jsonl 仅包含曾被 pooling/citation 命中的约 12K 文档，允许用于 smoke test，但正式 retrieval 必须回到 500K corpus。
- SciFact-Open pooling evidence 的 sentence highlights 是模型预测，不是全部人工标注，因此第一阶段不要求 sentence-level rationale F1；重点放在 document retrieval 与 document-level verification。
- 不要自行改标签定义或“修正”evaluator。所有主结果直接采用作者仓库 evaluator 输出的字段。

3. Baseline Selection

Priority	Method	Function	Why included	Phase-1 action
Must	MultiVerS	Claim-document verification + rationale	Task-native strong baseline; public SciFact checkpoint and inference code.	Run checkpoint inference only. No training.
Must	BM25 / simple sparse retrieval	Open-domain document retrieval	Cheap, interpretable, sufficient to validate SciFact-Open pipeline.	Index full 500K corpus and report Recall@K.
Should	VeriSci	Original SciFact full pipeline	Official repository has dev pipeline scripts and	Run only if legacy environment works

			pretrained models.	quickly.
Optional	ComboScorer / +VeriRel	Verification-aware reranking	Highly relevant to evidence retrieval, but official repo expects training V_MultiVerS and reports A100 training.	Do not train. Try only if usable pretrained outputs/checkpoint can be found quickly.
Later	FOLK / ProVe / P-Mem	Knowledge/provenance/memory comparisons	Structurally closer to knowledge management, but not necessary for fast dataset validation.	Defer.

Baseline principle 第一阶段必须有“一强一弱”两类结果：强 verifier = MultiVerS；廉价 retrieval = BM25/同等级 sparse retriever。VeriSci 用于补 full-pipeline 参照。+VeriRel 是研究参考，不应因为需要训练而阻塞阶段进度。

4. Required Working Directory and Logging Contract

Coding agent 应建立一个薄的 orchestration workspace，而不是把多个作者仓库改成一个工程。第三方代码保存在 external/ 并记录 commit；自己的 adapter、wrapper 和评测脚本放在 scripts/。

```

akb_baseline_phase1/
  README.md
  RUNLOG.md
  external/
    scifact/
    multivers/
    scifact-open/
    verirel/          # optional; clone only if attempted
  scripts/
    prepare_scifact_multivers.py
    build_scifact_open_index.py
    retrieve_scifact_open.py
    adapt_retrieval_to_multivers.py
    collect_metrics.py
  outputs/
    scifact/multivers/
    scifact/verisci/
    scifact_open/official_eval/
    scifact_open/bm25/
    scifact_open/multivers_topk/  # optional
  reports/
    baseline_results.md
    blockers.md
  env/
    environment_notes.md

```

4.1 RUNLOG.md minimum record

- Timestamp + task ID (e.g., SF-A2, SFO-B3).
- Repo commit hash(es).
- Exact command.
- Python / CUDA / PyTorch versions and device used.
- Input split and data path.

- Output path and wall-clock runtime.
- Status = PASS / FAIL / SKIP; FAIL must include the first actionable root cause, not the full raw traceback.

5. Execution Workflow

Order matters 严格按 A -> B -> C -> D 执行。不要一开始同时修多个 repo，也不要先做 +VeriRel。每个 work package 达到 exit condition 后立刻进入下一项。

A. Bootstrap and evaluator sanity (target: 1-1.5 h)

A1 Clone/pin repositories: Clone allenai/scifact, dwadden/multivers, dwadden/scifact-open. Record git rev-parse HEAD in env/environment_notes.md.

A2 Use repo-specific environments: SciFact recommends Python 3.7; MultiVerS and SciFact-Open recommend Python 3.8. Do not force all repos into one modern environment. Prefer separate conda envs or isolated containers.

A3 Download official data: Use repository scripts exactly once; do not manually rewrite dataset files. Preserve raw downloads.

A4 Check evaluator entry points: Confirm SciFact dev evaluator/pipeline can launch; confirm SciFact-Open python script/run_eval.py runs on provided model_predictions.

Exit condition: SciFact data are present; SciFact-Open run_eval.py prints metrics from author-provided predictions. If this fails because a download URL is dead, record blocker and try the official mirror/release only; do not substitute an unrelated repackaged dataset.

B. SciFact quick baseline validation (target: 2-3 h)

SF-B1 Official VeriSci pipeline smoke: Attempt the official dev full-pipeline path (open retrieval + verisci). The SciFact README exposes ./script/pipeline.sh [retrieval] [model] [dataset], with dev producing local metrics. If environment/model download becomes a long legacy-dependency repair, stop after the time cap and mark SKIP.

SF-B2 MultiVerS checkpoint setup: In the MultiVerS repo, install requirements in its own Python 3.8 env, download data with script/get_data.sh and the public scifact checkpoint using script/get_checkpoint.py scifact.

SF-B3 Run on public dev: Use MultiVerS prediction code against the public SciFact dev claims and corpus. If the convenience wrapper targets another split, call multivers/predict.py directly or generate a dev-format input file. Do not train.

SF-B4 Evaluate: Use the SciFact evaluator expected by MultiVerS / official SciFact format. Save raw predictions and evaluator stdout/JSON. Record all metrics emitted; at minimum preserve label-related and evidence-related scores.

SF-B5 One-line interpretation: Write whether the checkpoint works out-of-the-box, approximate runtime, and whether the output format can later be replaced by AKB-produced evidence/status decisions.

Exit condition: outputs/scifact/multivers/ contains predictions + evaluator output, and reports/baseline_results.md has one populated SciFact row. VeriSci is optional.

C. SciFact-Open full-corpus retrieval baseline (target: 2-3 h)

SFO-C1 Confirm official data: Run scifact-open script/get_data.sh. Verify data/claims.jsonl and the full data/corpus.jsonl (~500K abstracts). Also note corpus_candidates.jsonl separately as debug-only.

SFO-C2 Reproduce official evaluation: Run python script/run_eval.py and save output. This validates the evaluator before introducing new predictions.

SFO-C3 Build simple sparse index: Create a BM25-compatible index over title + abstract for all 500K corpus documents. Use Pyserini if installation is straightforward; otherwise implement a lightweight BM25/TF-IDF retriever. Prefer correctness and reproducibility over speed optimization.

SFO-C4 Retrieve top-K: For every SciFact-Open claim, retrieve at least top 100 document IDs. Save rankings with score and rank. Do not query only corpus_candidates for the formal run.

SFO-C5 Evaluate retrieval: Adapt rankings to the repository evaluator/prediction format. Report the retrieval metrics supported by the official evaluation code; always keep Recall@1/5/10/50/100 if they can be computed unambiguously from gold document evidence.

SFO-C6 Sanity sample: Inspect 5 random claims and verify that returned doc_ids exist in corpus.jsonl and that evidence doc_ids can be matched by the evaluator. This is a data-integrity check, not an error-analysis study.

Exit condition: outputs/scifact_open/bm25/ contains rankings + metrics generated from the full corpus. A smoke metric on ~12K candidate documents may be kept separately but must be labeled DEBUG.

D. Optional end-to-end verification on SciFact-Open (target: <= 1.5 h)

SFO-D1 Adapter: Convert each claim + retrieved top-K doc_ids into MultiVerS input schema. MultiVerS only verifies the documents listed in doc_ids, so retrieval is the gate between the 500K corpus and verifier.

SFO-D2 Small smoke first: Run 10-20 claims with K=5 or K=10. Confirm predictions serialize correctly and GPU memory/runtime are acceptable.

SFO-D3 Full run only if cheap: If the smoke run is stable, run all 279 claims. Otherwise stop and keep retrieval as the Phase-1 SciFact-Open result.

SFO-D4 Do not overclaim: SciFact-Open pooled sentence highlights are not all expert-annotated. Treat document-level evidence/label evaluation as primary; sentence-level rationale results are optional diagnostics only.

6. Failure Handling and Stop Rules

The coding agent should optimize for useful evidence of feasibility, not for rescuing every legacy repository.

- Hard time cap: one baseline/environment gets approximately half a focused day. If it still cannot enter inference/evaluation, stop and log the blocker.
- Dependency conflict: first try the repository-specified Python version. Do not mass-upgrade transformers/torch just to satisfy current defaults; this often breaks checkpoint compatibility.

- Checkpoint unavailable: search only author repository/release/model link. If no official checkpoint is available, do not substitute a retrained third-party checkpoint without marking it as non-official.
- OOM: reduce inference batch size to 1, reduce top-K for smoke runs, or move to CPU only to validate serialization. Do not change model architecture.
- Evaluator mismatch: adapt output schema, not evaluator logic. Never edit gold labels or scoring code to make predictions pass.
- SciFact-Open retrieval too slow: optimize indexing/querying, but do not shrink formal corpus. A reduced candidate-corpus run is allowed only as DEBUG.

7. Required Result Artifacts

7.1 baseline_results.md table

Dataset	Method	Split/Corpus	Checkpoint	Status	Metric(s)	Runtime	Notes
SciFact	MultiVerS	dev	scifact	PASS/FAIL	official evaluator output	xx min	
SciFact	VeriSci	dev/open	official	PASS/SKIP	official pipeline output	xx min	
SciFact-Open	Official stored predictions	279 claims	author-provided	PASS/FAIL	run_eval.py output	xx sec	evaluator sanity
SciFact-Open	BM25	500K corpus	N/A	PASS/FAIL	Recall@K / official retrieval metrics	xx min	formal retrieval
SciFact-Open	BM25 -> MultiVerS	top-K	scifact	OPTIONAL	document-level verification	xx min	only if cheap

7.2 Minimum files that must exist

- outputs/scifact/multivers/predictions.*
- outputs/scifact/multivers/metrics.*
- outputs/scifact_open/official_eval/metrics.txt
- outputs/scifact_open/bm25/rankings.*
- outputs/scifact_open/bm25/metrics.*
- RUNLOG.md
- reports/baseline_results.md
- reports/blockers.md (may be empty, but file should exist)

8. Phase-1 Completion Gate

When the required runs finish, do not immediately add more benchmarks. First answer the following four questions in reports/baseline_results.md:

1. Can SciFact be evaluated locally with a strong public verifier without training?
2. Can SciFact-Open be evaluated on the full 500K corpus with a reproducible retrieval baseline?
3. Can retrieval output be transformed into a verifier/AKB-compatible claim -> evidence-document interface with only a thin adapter?
4. Which baseline should remain as the fixed comparison when AKB is introduced: MultiVerS verifier, VeriSci full pipeline, BM25 retrieval, or a combination?

Phase-1 is DONE when SciFact-MultiVerS + SciFact-Open official-eval sanity + SciFact-Open full-corpus retrieval all have reproducible outputs. Everything else is optional.

9. Suggested Next Phase (do not execute yet)

- Replace the verifier / evidence-management stage with AKB output while keeping the same dataset and evaluator.
- Add a more knowledge-management-oriented comparison (e.g., provenance/evidence-aware system) only after the core metrics are stable.
- Then select one structurally different benchmark (MetaSyn or SciQA-ORKG) to test whether the interface generalizes.
- Finally use ResearchClawBench as an agent/harness integration experiment rather than as the first baseline environment.

10. Official Implementation Anchors

[SciFact official repository](#) - Author repo; README provides dev pipeline scripts and pretrained model downloads.

[MultiVerS official repository](#) - Author repo; provides checkpoints, inference code, dataset adapter format and SciFact model.

[SciFact-Open official repository](#) - Author repo; get_data.sh includes claims, predictions and full 500K corpus; run_eval.py reproduces paper metrics.

[+VeriRel official repository](#) - Useful research reference. Repo instructs training V_MultiVerS and reports single-A100 training, so it is optional in this phase.

Repository instructions and availability verified against official GitHub pages on 2026-08-08. If code changes after that date, prefer the current author README but preserve the commit hash used for each run.